

ASHIRBAD PATTNAIK

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PROFESSIONAL SUMMARY

Final-year Computer Science undergraduate (B.Tech, NIST University, Berhampur, CGPA 7.9) with hands-on, full-stack backend engineering experience: building and testing REST APIs (Node.js/Express, Python/FastAPI), implementing authentication and role-based access control, and containerizing multi-service applications with Docker Compose and CI/CD via GitHub Actions. Comfortable across the stack — from relational data modeling (SQL, SQLite, MySQL, Neo4j) to React/TypeScript frontends — with applied exposure to machine learning, explainable AI (XGBoost, SHAP), NLP, and LLM API integration through independent projects, including a full-stack AI/ML clinical decision-support platform with a RAG-based LLM component. Brings a QA-oriented, detail-driven working style shaped by a data-validation internship and disciplined algorithmic practice (LeetCode), plus experience translating undocumented enterprise ERP workflows into clear technical documentation during an industrial internship at RINL Visakhapatnam Steel Plant.

TECHNICAL SKILLS

Programming Languages: Python, Java (Core Java, OOP), JavaScript, TypeScript

Backend & APIs: Node.js, Express.js, FastAPI, REST API Design, Bcrypt Authentication, SQLAlchemy, J2EE

AI / Machine Learning: Machine Learning, XGBoost, SHAP (Explainable AI), NLP, LLM API Integration, RAG, Model Evaluation

Frontend: React, React Native, HTML5, CSS3, Tailwind CSS

Databases: SQL, MySQL, SQLite, DBMS, Database Design, Neo4j (Certified)

Cloud & DevOps: AWS (hands-on + Solutions Architecture Job Simulation, Forage), Docker & Docker Compose, GitHub Actions (CI/CD); learning GCP & Kubernetes

Tools & Version Control: Git, GitHub (branching, PRs, code review), VS Code, Jupyter Notebook, Postman

Core CS: Data Structures & Algorithms, Operating Systems, Computer Networks, Software Engineering / SDLC

AI-Assisted Development: Daily use of Claude and ChatGPT for coding, debugging, and prompt-based/agentic workflows

EXPERIENCE

IT & ERP Intern — RINL Visakhapatnam Steel Plant
Visakhapatnam, India

May 2026 – Jun 2026

- Contributed to live enterprise backend deliverables using Full Stack Development, J2EE, and Web 2.0 technologies in a deadline-driven production environment.
- Investigated undocumented ERP workflows and enterprise system behaviors, translating findings into clear technical write-ups and tasks for cross-functional IT staff.
- Supported ERP records administration and reporting accuracy alongside technical analysis, communicating progress consistently to stakeholders.

Data Science & Data Analysis Intern — NIST University
Berhampur, Odisha

May 2025 – Jun 2025

- Processed, cleaned, and validated structured datasets using Python (Pandas, Matplotlib), performing exploratory data analysis and visualization to surface actionable insights.
- Compiled and analyzed reports to support clear, evidence-based communication of technical findings to stakeholders.

Advanced Programming & Competitive Coding Intern — NIST University
Berhampur, Odisha

Jul 2024 – Aug 2024

- Solved algorithmic problems spanning trees, graphs, recursion, linked lists, stacks, and hash-based structures, reinforcing complexity analysis and edge-case awareness.

Data Structures & Algorithms Training — IT World Education
Berhampur, Odisha

Aug 2024 – Nov 2024

- Completed a 60-hour intensive DSA training program covering sorting algorithms, tree/graph traversal, and recursion-based problem solving, building a reference set of solved problems for technical interviews.

KEY PROJECTS

MediTwin AI — Explainable Clinical Decision Support Platform

Python, FastAPI, SQLAlchemy, XGBoost, SHAP, React, TypeScript, Docker Compose, GitHub Actions, LLM/RAG

- Built a REST API backend (FastAPI, SQLAlchemy) serving XGBoost-based ML predictions with a SHAP-based explainability layer, paired with a React/TypeScript frontend for interactive visualization.
- Integrated a RAG-based LLM component to support clinical decision context, and containerized/orchestrated the multi-service application (API, frontend, database) with Docker Compose and a GitHub Actions CI/CD pipeline.

RINL Material Gate Pass System

Node.js, Express.js, HTML5, Tailwind CSS, SQLite3, Bcrypt

- Designed and built REST-style API endpoints and relational data models with SQLite3 for persistence, to digitize and track material gate passes.
- Implemented Bcrypt-based authentication and role-based access control to restrict system access to authorized personnel.

ATS Resume Checker

Python, Streamlit, NLP, LLM API

- Built an NLP-powered application integrating a third-party LLM API to score resumes against job descriptions and surface keyword/skill gaps, deployed as an interactive Streamlit web tool.

INTERACT — Social Media Platform

React Native, AWS, JavaScript

- Developed a cross-platform mobile social app with backend services hosted on AWS, implementing core features (profiles, feed, interactions) end to end.

Menstruation Tracker

Python, SQLite/MySQL, React Native

- Designed a relational database schema for a mobile health-tracking app, focused on data privacy and efficient querying of historical records.

EDUCATION

B.Tech, Computer Science and Technology — NIST University

Expected 2027

Berhampur, Odisha — CGPA: 7.9 / 10.0

- Relevant coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Software Engineering, Object-Oriented Programming, Artificial Intelligence, Machine Learning, NLP.

Higher Secondary (Class XII), Science (PCMB) — 67.67% — Saraswati Vidya

2023

Mandir H.S. School

Secondary (Class X), General Studies — 91.33% — Saraswati Sishu Vidya Mandir

2021

CERTIFICATIONS & JOB SIMULATIONS

- AWS Solutions Architecture Job Simulation — AWS (Forage)
- Java Certification — One Roadmap
- AI Engineer Certification & AI and Data Scientist Certification — One Roadmap
- Neo4j Certified Professional
- Software Engineering Job Simulation — Electronic Arts (Forage)
- GenAI Job Simulation — Boston Consulting Group (Forage)
- GenAI-Powered Data Analytics Job Simulation — Tata (Forage)
- Data Analytics Job Simulation — Deloitte (Forage)

ADDITIONAL INFORMATION

Languages: English (C1), Hindi (C1), Odia (Native)

Activities: Brand Executive, LaunchEd Global; Participant, Trithon 2026 Hackathon

Awards: Matric Super Topper (Prameya); Certificate of Merit, UTSE 2020; Certificate of Appreciation, Bajaj Capital